European Union Dataset and Annotation Tool for Real Time Automatic License Plate Detection and Blurring

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***Abstract*— Automatic license plate detection has always been a popular topic in intelligent transport systems. In recent years, many approached the problem using artificial intelligence and machine learning techniques. Sufficient amount of good quality data is critical for machine learning, however, most of the existing open source license plate datasets are either restrictive in nature or do not have enough number of images to allow training a robust license plate detection network. In this paper, the THI License Plate Dataset (TLPD) is presented. It has more than 17,000 vehicle images and 18,000 labelled license plates in it. It is one of the largest publicly available European Union license plate datasets. The images in the dataset capture license plates at different angles and distances in relation to the camera. They are also taken under different illumination and weather conditions, making the dataset suitable for training robust license plate detectors that could work in various scenarios. A fast labelling tool for single class labelling is also presented in this paper. This tool was used to label the license plates in TLPD. This paper also introduces two license plate detection networks trained with data from TLPD. The networks work robustly under various scenarios, achieving precision and recall value of 93% under day and night conditions and 87% under heavy snowing condition.**

***Index Terms*— european license plate dataset, machine learning, license plate detector.**

I. INTRODUCTION

In recent years, many works have tried to do automatic license plate detection using artificial intelligence [1], [2], [3], [4], [5]. Although the efforts to solve the problem, license plate detection in uncontrolled and non-restrictive conditions remains a challenge, as most proposed methods have low accuracy when trying to detect license plates that are rotated, under uneven illumination, in snow weather or dark environment. All papers mentioned above trained and tested their detectors with very limited datasets (with only small number of unique images or small variation of angles), therefore only work under specific conditions.

Another problem with license plate datasets is that they are not universal. License plates from different countries can vary in color, size and characters (Chinese license plates have logograms on them). Z. Xu et al. [6] published the Chinese City Parking Dataset (CCPD) with more than 250,000 unique car images in 2018, which is a very sufficient amount of unique images to train a license plate detector, however, the dataset consists only of Chinese license plates, which have a blue background and Chinese characters (Fig. 1). If used to train a license plate detector, the detector might be biased and only work well on detecting Chinese license plates.

Fig. 1: Sample images from CCPD license plate dataset [6]

The motivation of this work comes form the national data protection law [7] that came into force in Europe in May 2018. Under the law, all personal items that can be used for identification must be anonymized in captured data. In Europe, license plate number is one type of data that could lead to identification. Therefore, a tool that could detect and blur license plates in videos and images is necessary.

The aims of our work were as follows: 1) To create an European Union License Plate Dataset which has images taken under day, night and snowing conditions, with variations in angles and distances of license plates in relation to the camera. 2) To capture and label a dataset big enough to train a license plate detector (with several thousand images). 3) To train a license plate detector that works in non-restricted environments.

II. RELATED WORK

This section is divided into two subsections. The first subsection will outline some existing license plate datasets from different countries and the second subsection will show some existing fast algorithms that could detect license plates in real time (FAST-ALPD).

A. Existing License Plate Datasets

Some of the existing license plate datasets have images captured from only one perspective in relation to the vehicle

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Fig. 2: Sample images from Reld dataset [8]

and most of them are not freely available for download. Reld [8] is a large license plate dataset with more than 182,000 license plate images. However, their images are extracted from videos taken by cameras placed on bridges, thus the license plates were only captured in some particular positions. Moreover, since the dataset was build for license plate recognition instead of detection, the dataset includes only images of license plates (Fig. 2), therefore they cannot be used to train license plate detection networks. The CCPD [6] has 250,000 images of cars parking in the street in China. The images were captured by cameras under day and night conditions. The UFPR dataset [4] and SSIG dataset [9] are two Brazilian license plate datasets. The images of these datasets were collected by cameras on sunny days. They have 4,500 and 2,000 license plates respectively and rarely have tilted license plates in them. Regarding European license plate datasets, a commonly used one is the openALPR dataset [10], but the dataset has only 97 license plates which is not enough to train a robust license plate detection neural network.

B. Existing Fast-ALPD Algorithms

Many automatic license plate detection algorithms were already developed. Traditional algorithms [11], [12], [13], [14] are normally based on image processing techniques. Early in 2008, Deb et al. [11] tried to exploit the fact that background of license plates usually exhibits a regular color. They used the HSI color model to detect license plate regions and achieved 89% accuracy but on a dataset with only 90 images. Later S. Azam and M. M. Islam [12] introduced a contrast enhancement method to locate license plates in low contrast indoor, night, blurry and foggy images. They achieved 98.15% detection rate on a dataset of 325 English car images and 91.24% detection rate on a dataset of 525 Bangladeshi car images. S. Yu et al. [13] proposed to use wavelet transform and empirical mode decomposition analysis to detect license plates and achieved 97.91 detection rate on a dataset of 765 images, with 16 of them taken at night; while A. K. Bachchan [14] suggested the method of local binary pattern and histogram matching and achieved 96.14% detection accuracy on a dataset of 300 images. In recent years Region-based Convolutional Neural Networks (R-CNN) are being used to develop more robust license plate detectors. Z. Xu el al. [] M. Dong et al. [15] built a dataset with 18,699 car images with Chinese license plates

Fig. 3: Sample images of German Motorcycle license plate (some characters in the image are blocked due to privacy protection)

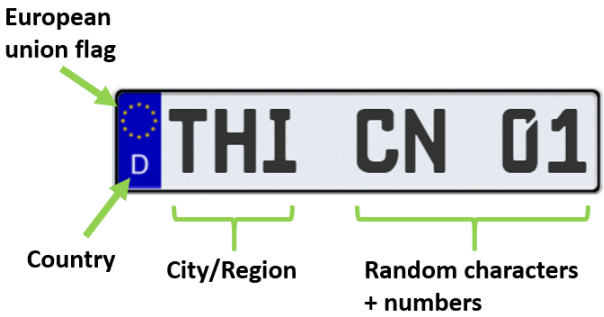

Fig. 4: Format of a typical German passenger vehicle license plate

in good weather and daylight conditions. On their work, a R-CNN was trained and achieved 96.68% precision at 50% intersection over union (IoU). The You Only Look Once (Yolo) neural network was published by J. Redmon et al. [16] in 2016 and two other improved versions in 2017 [17] and 2018 [18]. Many works [3], [19], [4], [5], [20] used different variations of the Yolo network to train license plate detectors and achieved good results of over 90% precision and recall at 50% IoU.

III. Methodology

This section is divided into two subsections. The first subsection will desribe the data acquisition process of the THI dataset. The second section will describe the training process of the license plate detectors.

A. Data Acquisition

All the images from the THI dataset were extracted from video sequences. The first stage of data acquisition was to take videos of vehicles on roads and in car parking places in Germany and France. The iPhone XS MAX and the Asus Zenfone were used to record the videos. These videos were taken during different times of the day and diverse weather conditions, including snow. The camera was rotated and shifted upwards and downwards in relation to the vehicles to record different angles of license plates. Additional videos of motorcycles license plates were also added to the dataset (Fig. 3), since they have different shapes and colors when compared to license plates of normal cars (Fig. 4).

The videos were shot with a 30 frames per second sampling rate and an HD resolution of 1,980 x 1,020 pixels. To make sure that the extracted images were not too similar, one

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TABLE I: License Plate Detector Neural Network Training Parameters

	Fast-Yolo	Tiny-Yolov3
Batch size	128	512
Image size	416 x 416	608 x 608
Momentum	0.9	0.9
Decay	0.0005	0.0005
Learning rate	0.0001	0.001
No. of anchor boxes	5	6
No. of epochs	30,000	30,000

image from every 10 frames were selected from the videos to form the THI License Plate Dataset (TLPD). An open source fast labelling tool called FastLabel was developed to process the images. This tool allows us to draw bounding boxes on top of the area to be labelled. More details of FastLabel is reported in section IV-A. All the license plates in the dataset were manually labelled using this tool.

As some previous papers used cropped car images as input for their detector [4], [3], cropped car images with labelled license plates were made available in TLPD. A Yolov3 detector [18] was used to produce the cropped car images. Among the 80 classes that Yolov3 detector can detect, car, motorcycle, bus and truck were chosen to be cropped out from the dataset. After cropping, a check was made to guarantee the presence of at least one license plate in the cropped image, generating a label for it and saving it along the image in the dataset.

B. Training the License Plate Detectors

Two license plate detectors were trained using the TLPD cropped image set. To train the detectors, the dataset was divided into 60% for training, 15% for validation and 25% for testing. To avoid similar frame sequences in both datasets that may lead to artificially good classification results, only images from different video sequences were used for the training, validation and testing data.

The first detector used transfer learning on top of the Fast-Yolo architecture, which is the same used in the UFPR paper [4] and the second one, over the Tiny-Yolov3 architecture [18]. Each one was trained during 30,000 epochs. The training was done in a computer with GeForce RTX2080Ti graphic card. The training parameters are shown on Table I.

All the parameters for training the Fast-Yolo network were the same as the parameters used by R. Laroca et al. [4]. For Tiny-Yolo v3 architecture, the batch size was increased to 512, with a larger batch size, the learning rate was increased to 0.0001, image size and number of anchor boxes were changed to 608 x 608 and 6 respectively. Fine tuning was done after transfer learning with the detectors, the initial weights for the Fast-Yolo architecture are from the UFPR license plate detector network [4], while the initial weights for the Tiny-Yolov3 architecture are from the original Tiny-Yolov3 object detector [18].

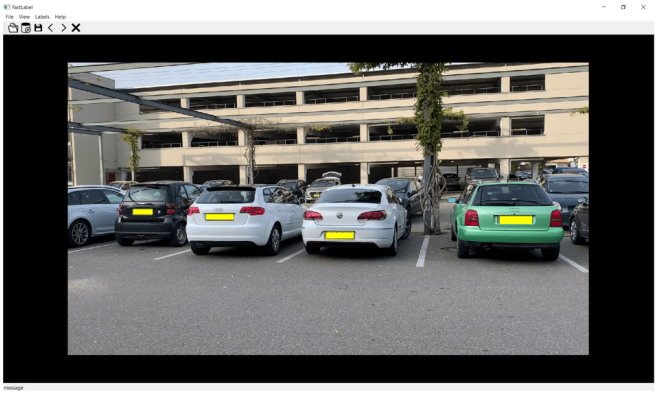

Fig. 5: FastLabel user interface

IV. Experimental Results

A. Labelling Tool: FastLabel

FastLabel is designed to label one class of object in a dataset at a time, such as license plates, for example. All the labels' description are saved in one text file. The name of the text file indicates the class name. The descriptions of a label include the x and y coordinates of the top left corner, the width and the height of the bounding box. Since the user is not required to assign a class for each label, the time for labelling one class of objects in a dataset is significantly reduced. FastLabel also allows user to zoom in and out with the mouse scroll wheel and shift view along the image with right mouse button while labelling. The user interface is also very simple 5, making FastLabel a very suitable tool for labelling dataset with only a few detection classes. FastLabel was made freely available for download together with TLPD.

B. TLPD

Our dataset contains images of cars, trucks and motorcycles taken under different environments, illuminations, pose and occlusion. TLPD has two sub-sets, the HD image set which contains 6,088 HD images of the environment, normally with more than one license plate within, and the cropped image set that contains 17,105 cropped images of cars, trucks and motorcycles of different sizes.

The images in the HD image set were extracted from 43 different videos of 30 fps by taking every 10th frame from the videos, recorded under day, night and snow conditions (see Fig. 6). The dataset includes images taken under different weather and light conditions in order to increase the robustness of a detector trained with those images.

The cropped image set was generated from the HD image set using Yolo v3 detector. Images of the HD image set were fed to Yolo v3 detector to detect for vehicles. Once detected, if one or more license plates were present in the detected region, the image of the vehicle was saved together with the description of the bounding box of the license plate. Some images from the TLPD cropped set are shown in Fig 7.

Comparing to the commonly used datasets reviewed before, TLPD is the only dataset that has 1) both HD and

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TABLE II: A comparison of publicly available license plate datasets and TLPD

	Reld	UFPR	SSIG	CCPD	Open-ALPR(EU)	TLPD (THI)
Number of images	182,000	4,500	2,000	250,000	97	17,105
Number of LPs	182,000	4,500	2,000	250,000	97	18,672
Country / Region	Czech Republic	Brazil	Brazil	China	European	Germany + France
Full HD images available	X	✓	✓	X	X	✓
Cropped vehicle image	X	X	X	✓	✓	✓
Variation in tilt degrees	X	X	X	✓	✓	✓
Variation in illumination	X	X	X	✓	X	✓
Images in snowing weather	X	X	X	X	X	✓
Significant amount of motorcycles	X	X	X	X	X	✓

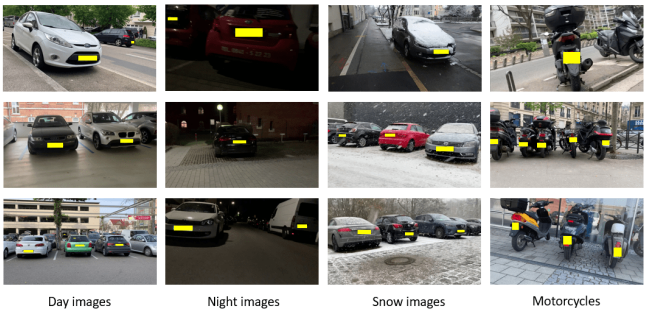

Fig. 6: Images from TLPD (license plates are blocked due to privacy protection)

Fig. 7: Images from TLPD cropped image set (license plates are blocked due to privacy protection)

cropped images, 2) variations in different tilt degrees, 3) variation in different illumination levels, 4) images in snowing weather and 5) more than 500 labelled motorcycle license plates. More details are reported in Table II.

Table III shows the total number of images and labels in the HD image set and cropped image set of TLPD. It also shows the percentage of each category of image in the dataset. The HD image set has 6,088 images and 13,761 labels, while the cropped set has 17,105 images and 18,672 labels. The cropped image set has more labels than the HD image set because a license plate could appear in more than one cropped image, thus the number of labels is bigger after images are cropped from the HD image set to form the cropped image set.

TABLE III: Distribution of Images from different Conditions in TLPD

	HD image set		Cropped image set	
	images	labels	images	labels
Day	3,134 (51%)	7,870 (57%)	10,198(60%)	10,947 (59%)
Night	1,674 (27%)	2,940 (21%)	3,348(20%)	3,685 (20%)
Snow	1,002 (16%)	2,287 (17%)	2,707(16%)	3,088 (17%)
Total	6,088	13,761	17,105	18,672

TABLE IV: Precision and Recall of License Plate Detection Network at 50% IoU

	precision	Recall
Fast-Yolo trained on real data	90.08%	90.11%
Tiny-Yolo v3 trained on real data	91.44%	92.32%

TABLE V: Precision and Recall of Tiny-Yolo v3 detector at 50% IoU

	Day	Night	Snow
Precision at 50% IoU	92.74%	93.84%	86.97%
Recall at 50% IoU	93.17%	93.94%	87.65%

C. Performance of the LP detectors

The performance of the license plate detection networks were evaluated by their precision and recall values on the test set (Table. IV). The license plate detection network trained with Fast-Yolo architecture achieved precision and recall values at 50% IoU (intersection over union) of 91.9% and 92.11% on the test set respectively; while the license plate detection network trained with Tiny-Yolo v3 architecture achieved precision and recall values at 50% IoU of 92.97% and 93.32% on the test set respectively. The Tiny-Yolo v3 license plate detector outperformed the Fast-Yolo v3 detector.

The performance of the Tiny-Yolo v3 detector trained with data at day, night and snowing conditions were evaluated individually. Table V shows the precision and recall of the Tiny-Yolo v3 detector at 50% IoU. The precision and recall of the Tiny-Yolo v3 detector at daytime and night time are around 93%, but in snowing conditions the values drop to 87%. This result shows that the network is equally accurate under daytime and night-time condition, but less accurate in snowing conditions.

D. Capability of the Tiny-Yolo v3 license plate detector

As it can be seen in Fig. 8 and 9, the license plates in the images of the testing set were successfully detected. The blue bounding box indicates the ground truth position of the license plate while the red box indicates the result of the detection. An analysis of the results on the testing

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Fig. 8: License plates under day conditions successfully detected in test set (license plates are blocked due to privacy protection)

Fig. 9: License plates under night and snow conditions successfully detected in test set (license plates are blocked due to privacy protection)

set indicates that the detector network performs successfully under the following situations:

1) Plates on front and back of a vehicle
2) Plates at different distances
3) Plates at different angles
4) Plates on different types of vehicles
5) Day and night conditions
6) Plates with at least of 25% visible area
7) Blurry plates

The Tiny-Yolo v3 network was tested with an NVIDIA RTX 2080 graphic card and it was able to process videos with a resolution of 1,980 x 1,020 pixels at 30 frames per second.

V. Conclusions

This paper presented TLPD, one of the largest open source European Union license plate datasets currently available, and FastLabel, a fast and convenient tool for labelling datasets. They are available to download by the research community as indicated in VI.

Two license plate detection networks were trained with data from the TLPD using transfer learning and fine tuning over Fast-Yolo and Tiny-Yolov3 architectures. The performance of the detectors was satisfactory, with both achieving more than 90% precision and recall on test sets where the images were taken from videos never exposed to the network, recorded at completely different places. The testing result shows that the networks are able to detect license plates under different non-restrictive environments with 30 frames per seconds, thus proving that the TLPD dataset is suitable for developing a robust license plate detector able to blur plates for European Union data protection law commpliance.

VI. To Download the Dataset and FastLabel

TLPD and FastLabel are available in the following link: www.thi.de/go/license-plate-detection. Please read the related terms and conditions before downloading the dataset. If you intend to use the dataset or the annotation tool in any context, please cite this paper.

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